

Safety data sheet

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BASF Safety data sheet according to the United Nations' Globally Harmonized System (UN GHS)

Date / Revised: 16.10.2025

Version: 5.0

Product: **Hydralan® 400 N.V.-1**

(ID no. 30053073/SDS_GEN_00/EN)

Date of print 05.12.2025

1. Identification

Product identifier

Hydralan® 400 N.V.-1

Relevant identified uses of the substance or mixture and uses advised against

Relevant identified uses: brake fluid

Details of the supplier of the safety data sheet

Company:

BASF SE

67056 Ludwigshafen

GERMANY

Fuel and Lubricant Solutions

Telephone: +49 621 60-0

E-mail address: RegXcellenceFuelLubes@basf.com

Emergency telephone number

International emergency number:

Telephone: +49 180 2273-112

2. Hazards Identification

Classification of the substance or mixture

According to UN GHS criteria

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Eye Dam./Irrit. 2A
Repr. 2 (unborn child)
Repr. 1B (fertility)

For the classifications not written out in full in this section the full text can be found in section 16.

Label elements

Globally Harmonized System (GHS)

Pictogram:



Signal Word:

Danger

Hazard Statement:

H319	Causes serious eye irritation.
H360	May damage fertility. Suspected of damaging the unborn child.

Precautionary Statements (Prevention):

P280	Wear protective gloves, protective clothing and eye protection or face protection.
P280	Wear eye protection.
P201	Obtain special instructions before use.
P202	Do not handle until all safety precautions have been read and understood.
P264	Wash contaminated body parts thoroughly after handling.

Precautionary Statements (Response):

P305 + P351 + P338	IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.
P308 + P313	IF exposed or concerned: Get medical attention.
P337 + P313	If eye irritation persists: Get medical attention.

Precautionary Statements (Storage):

P405	Store locked up.
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Precautionary Statements (Disposal):

P501	Dispose of contents and container to hazardous or special waste collection point.
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Labeling of special preparations (GHS):

May produce an allergic reaction. Contains: 4,4'-Isopropylidenediphenol

Other hazards

According to UN GHS criteria

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If applicable information is provided in this section on other hazards which do not result in classification but which may contribute to the overall hazards of the substance or mixture.

The product does not contain a substance above legal limits fulfilling the PBT (persistent/bioaccumulative/toxic) criteria or the vPvB (very persistent/very bioaccumulative) criteria.

3. Composition/Information on Ingredients

Substances

Not applicable

Mixtures

Chemical nature

Blend based on: polyglycol, glycol ether, inhibitors, glycoletherborate

Hazardous ingredients (GHS)

According to UN GHS criteria

Tris[2-[2-(2-methoxyethoxy)ethoxy]ethyl] orthoborate

Content (W/W): $\geq 40\%$ - $\leq 50\%$	Repr. 2 (fertility)
CAS Number: 30989-05-0	Repr. 2 (unborn child)
EC-Number: 250-418-4	H361

2-(2-(2-Butoxyethoxy)ethoxy)ethanol

Content (W/W): $> 10\%$ - $< 20\%$	Acute Tox. 5 (dermal)
CAS Number: 143-22-6	Eye Dam. 1
EC-Number: 205-592-6	H318, H313
INDEX-Number: 603-183-00-0	

Specific concentration limit:

Eye Irrit. 2: 20% - $< 30\%$
Eye Dam. 1: $\geq 30\%$

2-(2-Methoxyethoxy)ethanol

Content (W/W): $> 1\%$ - $< 3\%$	Flam. Liq. 4
CAS Number: 111-77-3	Repr. 1B (unborn child)
EC-Number: 203-906-6	H227, H360
INDEX-Number: 603-107-00-6	

Specific concentration limit:

Repr. 1B, unborn child: $\geq 3\%$

2,2'-Oxydiethanol

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Content (W/W): > 3 % - < 5 %
 CAS Number: 111-46-6
 EC-Number: 203-872-2
 INDEX-Number: 603-140-00-6

Acute Tox. 4 (oral)
 H302

1,1'-Iminodipropyl-2-ol

Content (W/W): < 3 %
 CAS Number: 110-97-4
 EC-Number: 203-820-9
 INDEX-Number: 603-083-00-7

Eye Irrit. 2A
 H319

4,4'-Isopropylidenediphenol

Content (W/W): > 0,1 % - < 0,5 %
 CAS Number: 80-05-7
 EC-Number: 201-245-8
 REACH registration number: 01-2119457856-23
 INDEX-Number: 604-030-00-0

Acute Tox. 5 (oral)
 Acute Tox. 5 (dermal)
 Eye Dam. 1
 Skin Sens. 1
 Repr. 1B (fertility)
 STOT SE 3 (irr. to respiratory syst.)
 Aquatic Acute 2
 Aquatic Chronic 2
 H318, H303, H317, H335, H360, H401, H411,
 H313

3,6,9,12-Tetraoxahexadecan-1-ol

Content (W/W): $\geq 2\%$ - $\leq 10\%$
 CAS Number: 1559-34-8
 EC-Number: 216-322-1

Acute Tox. 5 (oral)
 Eye Irrit. 2A
 H319, H303

For the classifications not written out in full in this section the full text can be found in section 16.

4. First-Aid Measures

Description of first aid measures

Immediately remove contaminated clothing. If the patient is likely to become unconscious, place and transport in stable sideways position (recovery position). First aid personnel should pay attention to their own safety.

If inhaled:

Keep patient calm, remove to fresh air, seek medical attention.

On skin contact:

Wash thoroughly with soap and water

On contact with eyes:

Wash affected eyes for at least 15 minutes under running water with eyelids held open, consult an eye specialist.

On ingestion:

Immediately rinse mouth and then drink 200-300 ml of water, seek medical attention.

Most important symptoms and effects, both acute and delayed

Symptoms: Information, i.e. additional information on symptoms and effects may be included in the GHS labeling phrases available in Section 2 and in the Toxicological assessments available in Section 11., (Further) symptoms and / or effects are not known so far

Indication of any immediate medical attention and special treatment needed

Treatment: Treat according to symptoms (decontamination, vital functions), no known specific antidote.

5. Fire-Fighting Measures**Extinguishing media**

Suitable extinguishing media:
water spray, dry powder, foam

Special hazards arising from the substance or mixture

harmful vapours

Generation of fumes/fog. The substances/groups of substances mentioned can be released in case of fire.

Advice for fire-fighters

Special protective equipment:
Wear a self-contained breathing apparatus.

Further information:

The degree of risk is governed by the burning substance and the fire conditions. Contaminated extinguishing water must be disposed of in accordance with official regulations.

6. Accidental Release Measures**Personal precautions, protective equipment and emergency procedures**

Use personal protective clothing. Breathing protection required.

Environmental precautions

Contain contaminated water/firefighting water. Do not discharge into drains/surface waters/groundwater.

Methods and material for containment and cleaning up

For large amounts: Pump off product.

For residues: Pick up with suitable absorbent material. Dispose of absorbed material in accordance with regulations.

7. Handling and Storage

Precautions for safe handling

No special measures necessary provided product is used correctly.

Protection against fire and explosion:

Take precautionary measures against static discharges.

Conditions for safe storage, including any incompatibilities

Unsuitable materials for containers: Paper/Fibreboard

Further information on storage conditions: Keep container tightly closed and dry; store in a cool place.

8. Exposure Controls/Personal Protection

Control parameters

Components with occupational exposure limits

80-05-7: 4,4'-Isopropylidenediphenol

110-97-4: 1,1'-Iminodipropyl-2-ol

111-46-6: 2,2'-Oxydiethanol

111-77-3: 2-(2-Methoxyethoxy)ethanol

112-35-6: 2-(2-(2-Methoxyethoxy)ethoxy)ethanol

30989-05-0: Tris[2-[2-(2-methoxyethoxy)ethoxy]ethyl] orthoborate

Exposure controls

Personal protective equipment

Respiratory protection:

Respiratory protection in case of vapour/aerosol release. Particle filter with medium efficiency for solid and liquid particles (e.g. EN 143 or 149, Type P2 or FFP2)

Hand protection:

Wear chemically resistant gloves in combination with specific activity training

Chemical resistant protective gloves (EN ISO 374-1)

Suitable materials for short-term contact (recommended: At least protective index 2, corresponding > 30 minutes of permeation time according to EN ISO 374-1)

butyl rubber (butyl) - 0.7 mm coating thickness

nitrile rubber (NBR) - 0.4 mm coating thickness

Supplementary note: The specifications are based on tests, literature data and information of glove manufacturers or are derived from similar substances by analogy. Due to many conditions (e.g. temperature) it must be considered, that the practical usage of a chemical-protective glove in practice may be much shorter than the permeation time determined through testing.

Manufacturer's directions for use should be observed because of great diversity of types.

Eye protection:

Tightly fitting safety goggles (splash goggles) (e.g. EN 166)

Body protection:

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chemical-protection suit (f.e. according to EN 14605)

General safety and hygiene measures

Wearing of closed work clothing is required additionally to the stated personal protection equipment.

No eating, drinking, smoking or tobacco use at the place of work. Handle in accordance with good industrial hygiene and safety practice.

9. Physical and Chemical Properties**9.1. Information on basic physical and chemical properties**

State of matter:	liquid	
Form:	liquid	
Colour:	yellow	
Odour:	product specific	
Odour threshold:	not determined	
solidification temperature:	< -50 °C	(DIN 51583)
boiling temperature:	> 260 °C	(ASTM D1120)
Flammability:	hardly combustible	(derived from flash point)
Lower explosion limit:	For liquids not relevant for classification and labelling., The lower explosion point may be 5 - 15 °C below the flash point.	
Upper explosion limit:	For liquids not relevant for classification and labelling.	
Flash point:	> 130 °C	(DIN 51758)
Auto-ignition temperature:	not determined	(DIN 51794)
Thermal decomposition:	No decomposition if correctly stored and handled.	
pH value:	7,5 - 9	(FMVSS 116, S 6.4)
Viscosity, kinematic:	14 mm ² /s (20 °C)	(DIN 51562)
Thixotropy:	not thixotropic	
Solubility in water:	fully soluble	
Solubility (quantitative) solvent(s):	polar solvents soluble	
Partitioning coefficient n-octanol/water (log Kow):	Study scientifically not justified.	
Vapour pressure:	1 mbar (20 °C) 1 mbar (50 °C)	
Density:	approx. 1,06 g/cm ³ (20 °C)	
Relative vapour density (air):	> 1 (20 °C) Heavier than air.	(estimated)

Particle characteristics

Particle size distribution: The substance / product is marketed or used in a non solid or granular form. -

9.2. Other information

Information with regard to physical hazard classes

Explosives

Explosion hazard: not explosive

Oxidizing properties

Fire promoting properties: not fire-propagating

Corrosion to metals

No corrosive effect on metal.

Other safety characteristics

Other Information: If necessary, information on other physical and chemical parameters is indicated in this section.

Evaporation rate: not determined

10. Stability and Reactivity

Reactivity

Corrosion to metals: No corrosive effect on metal.

Chemical stability

The product is stable if stored and handled as prescribed/indicated.

Possibility of hazardous reactions

No hazardous reactions when stored and handled according to instructions.

Conditions to avoid

Avoid open flames. Avoid humidity.

Incompatible materials

Substances to avoid:
strong oxidizing agents, strong bases, strong acids

Hazardous decomposition products

Hazardous decomposition products:
No hazardous decomposition products known.

11. Toxicological Information

Information on toxicological effects

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Irritation

Assessment of irritating effects:

Eye contact causes irritation. Not irritating to the skin.

Respiratory/Skin sensitization

Assessment of sensitization:

A sensitizing effect on particularly sensitive individuals cannot be excluded.

Germ cell mutagenicity

Assessment of mutagenicity:

Based on available data, the classification criteria are not met.

Carcinogenicity

Assessment of carcinogenicity:

The whole of the information assessable provides no indication of a carcinogenic effect.

Reproductive toxicity

Assessment of reproduction toxicity:

May impair fertility.

*Information on: 4,4'-Isopropylidenediphenol**Assessment of reproduction toxicity:**The results of animal studies suggest a fertility impairing effect.*

Developmental toxicity

Assessment of teratogenicity:

May cause harm to the unborn child.

*Information on: 2-(2-Methoxyethoxy)ethanol**Assessment of teratogenicity:**In animal studies the substance caused malformations.*

Specific target organ toxicity (single exposure)

Remarks: Based on available data, the classification criteria are not met.

Repeated dose toxicity and Specific target organ toxicity (repeated exposure)

Assessment of repeated dose toxicity:

Based on available data, the classification criteria are not met.

Aspiration hazard

No aspiration hazard expected.

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Other relevant toxicity information

The product has not been tested. The statement has been derived from the properties of the individual components.

12. Ecological Information

Toxicity

Assessment of aquatic toxicity:

There is a high probability that the product is not acutely harmful to aquatic organisms.

Persistence and degradability

Elimination information:

> 70 % DOC reduction (28 d) (OECD 302B; ISO 9888; 88/302/EWG, Teil C) Easily eliminated from water.

Bioaccumulative potential

Assessment bioaccumulation potential:

Significant accumulation in organisms is not to be expected.

Mobility in soil

Assessment transport between environmental compartments:

Volatility: The substance will not evaporate into the atmosphere from the water surface.

Results of PBT and vPvB assessment

According to Annex XIII of Regulation (EC) No.1907/2006 concerning the Registration, Evaluation, Authorisation and Restriction of Chemicals (REACH): The product does not contain a substance fulfilling the PBT (persistent/bioaccumulative/toxic) criteria or the vPvB (very persistent/very bioaccumulative) criteria.

Other adverse effects

The product does not contain substances that are listed in Regulation (EU) 2024/590 on substances that deplete the ozone layer.

13. Disposal Considerations

Waste treatment methods

Must be disposed of or incinerated in accordance with local regulations.

Contaminated packaging:

Uncontaminated packaging can be re-used.

Packs that cannot be cleaned should be disposed of in the same manner as the contents.

14. Transport Information

Land transport

ADR

Not classified as a dangerous good under transport regulations

UN number or ID number: Not applicable

UN proper shipping name: Not applicable

Transport hazard class(es): Not applicable

Packing group: Not applicable

Environmental hazards: Not applicable

Special precautions for user: None known

RID

Not classified as a dangerous good under transport regulations

UN number or ID number: Not applicable

UN proper shipping name: Not applicable

Transport hazard class(es): Not applicable

Packing group: Not applicable

Environmental hazards: Not applicable

Special precautions for user: None known

Inland waterway transport

ADN

Not classified as a dangerous good under transport regulations

UN number or ID number: Not applicable

UN proper shipping name: Not applicable

Transport hazard class(es): Not applicable

Packing group: Not applicable

Environmental hazards: Not applicable

Special precautions for user: None known

Transport in inland waterway vessel

Not evaluated

Sea transport

IMDG

Not classified as a dangerous good under transport regulations

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UN number or ID number:	Not applicable
UN proper shipping name:	Not applicable
Transport hazard class(es):	Not applicable
Packing group:	Not applicable
Environmental hazards:	Not applicable
Special precautions for user	None known

Air transport

IATA/ICAO

	Not classified as a dangerous good under transport regulations
UN number or ID number	Not applicable
UN proper shipping name:	Not applicable
Transport hazard class(es):	Not applicable
Packing group:	Not applicable
Environmental hazards:	Not applicable
Special precautions for user	None known

Maritime transport in bulk according to IMO instruments

Maritime transport in bulk is not intended.

15. Regulatory Information**Safety, health and environmental regulations/legislation specific for the substance or mixture**

Not applicable

16. Other Information

Full text of classifications, hazard symbols and hazard statements, if mentioned in section 2 or 3:

Eye Dam./Irrit.	Serious eye damage/eye irritation
Repr.	Reproductive toxicity
Acute Tox.	Acute toxicity
Eye Dam.	Serious eye damage
Flam. Liq.	Flammable liquids
Eye Irrit.	Eye irritation
Skin Sens.	Skin sensitization
STOT SE	Specific target organ toxicity — single exposure
Aquatic Acute	Hazardous to the aquatic environment - acute
Aquatic Chronic	Hazardous to the aquatic environment - chronic
H361	Suspected of damaging fertility. Suspected of damaging the unborn child.
H318	Causes serious eye damage.
H313	May be harmful in contact with skin.
H227	Combustible liquid.
H360	May damage the unborn child.
H302	Harmful if swallowed.

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H319	Causes serious eye irritation.
H303	May be harmful if swallowed.
H317	May cause an allergic skin reaction.
H335	May cause respiratory irritation.
H360	May damage fertility.
H401	Toxic to aquatic life.
H411	Toxic to aquatic life with long lasting effects.
H313	May be harmful in contact with skin.

The data contained in this safety data sheet are based on our current knowledge and experience and describe the product only with regard to safety requirements. This safety data sheet is neither a Certificate of Analysis (CoA) nor technical data sheet and shall not be mistaken for a specification agreement. Identified uses in this safety data sheet do neither represent an agreement on the corresponding contractual quality of the substance/mixture nor a contractually designated use. It is the responsibility of the recipient of the product to ensure any proprietary rights and existing laws and legislation are observed.

Vertical lines in the left hand margin indicate an amendment from the previous version.